

ADMET Properties of Top 10 SARS-CoV-2 Spike Protein Inhibitors

Predicted using ADMETlab 3.0 | Dimple Srivastava | Jamia Hamdard, Department of Virology

Table 1: Absorption and Distribution Properties

Compound	Target (PDB)	Docking Score (kcal/mol)	GI Absorption	Bioavailability Score	BBB Permeant	P-gp Substrate	VD (L/kg)	Plasma Protein Binding (%)
Hesperidin	6VXX	-10.41	Low	0.17	No	Yes	0.598	81.4%
Lumacaftor	6M0J	-8.60	High	0.56	No	Yes	0.30	98.7%
Oxyberberine	7DWZ	-8.58	High	0.55	Yes	No	1.033	93.8%
Mevastatin	7DWZ	-8.16	High	0.55	Yes	No	0.905	76.1%
Celecoxib	6VXX	-8.34	High	0.55	No	No	2.439	96.7%
Brazilin	6VXX	-8.34	High	0.55	No	Yes	1.173	86.2%
Andrographolide	7DWZ	-8.06	High	0.55	No	Yes	0.506	66.2%
SSAA09E3	6M0J	-7.89	High	0.55	Yes	No	0.746	97.0%
Meloxicam	6VXX	-8.23	High	0.56	No	No	0.232	99.5%
Indomethacin	6VXX	-7.84	High	0.85	Yes	No	0.116	98.4%

Table 2: Metabolism — CYP Enzyme Inhibition Profile

Compound	Target	CYP1A2	CYP2C19	CYP2C9	CYP2D6	CYP3A4	CL _{plasma}	T _½ (h)
Hesperidin	6VXX	0-0.1	0-0.1	0-0.1	0-0.1	0-0.1	1.433	4.918
Lumacaftor	6M0J	0.9-1.0	0.5-0.7	0.9-1.0	0-0.1	0-0.1	0.302	1.184
Oxyberberine	7DWZ	0.9-1.0	0-0.1	0.7-0.9	0.1-0.3	0.9-1.0	3.741	1.042
Mevastatin	7DWZ	0-0.1	0-0.1	0-0.1	0-0.1	0.9-1.0	9.173	1.062
Celecoxib	6VXX	0.7-0.9	0.7-0.9	0.9-1.0	0-0.1	0-0.1	3.78	0.651
Brazilin	6VXX	0.5-0.7	0-0.1	0-0.1	0-0.1	0.7-0.9	11.604	2.382
Andrographolide	7DWZ	0-0.1	0-0.1	0-0.1	0-0.1	0-0.1	9.674	1.567
SSAA09E3	6M0J	0.9-1.0	0.3-0.5	0.3-0.5	0.9-1.0	0.5-0.7	1.948	1.102
Meloxicam	6VXX	0-0.1	0.9-1.0	0.9-1.0	0-0.1	0.9-1.0	-0.059	1.077
Indomethacin	6VXX	0.1-0.3	0-0.1	0.9-1.0	0-0.1	0-0.1	0.845	1.484

Table 3: Toxicity Profile

Compound	Target	AMES Toxicity	hERG Blocker	Hepato- toxicity	Skin Sensitization	Rat Oral Acute Tox	Carcino- genicity	Overall Safety
Hesperidin	6VXX	High (0.944)	Low (0.030)	High (0.631)	High (0.983)	Low (0.024)	Low (0.034)	✗ Concern
Lumacaftor	6M0J	Low (0.148)	Low (0.453)	Low (0.211)	Low (0.022)	High (0.880)	Low (0.238)	~ Moderate
Oxyberberine	7DWZ	High (0.819)	Low (0.402)	High (0.702)	Low (0.256)	High (0.536)	High (0.917)	✗ Concern
Mevastatin	7DWZ	High (0.667)	Low (0.019)	High (0.668)	High (1.000)	Low (0.090)	High (0.509)	✗ Concern
Celecoxib	6VXX	Low (0.306)	Low (0.351)	High (0.965)	Low (0.004)	High (0.627)	Low (0.372)	~ Moderate
Brazilin	6VXX	Low (0.430)	Low (0.108)	High (0.790)	High (0.976)	Low (0.416)	High (0.889)	✗ Concern
Andrographolide	7DWZ	High (0.697)	Low (0.051)	High (0.850)	High (0.997)	High (0.848)	High (0.984)	✗ Concern
SSAA09E3	6M0J	High (0.889)	High (0.605)	High (0.762)	High (0.715)	Low (0.339)	High (0.585)	✗ Concern
Meloxicam	6VXX	Low (0.050)	Low (0.005)	High (0.526)	High (0.845)	High (0.976)	Low (0.034)	✗ Concern
Indomethacin	6VXX	Low (0.274)	Low (0.239)	High (0.881)	High (0.755)	High (0.954)	High (0.748)	✗ Concern

Note: Probability values ≥ 0.5 indicate positive prediction (High risk). Green = favorable | Red = concern | Amber = moderate. CYP inhibition ranges: 0-0.1 = No inhibition, 0.9-1.0 = Strong inhibition. Source: ADMETlab 3.0